For each of the following Chain rule problems, find the derivative of $f(x)$ using the form $\frac{df}{dx} = \frac{df}{du}\frac{du}{dx}$.

1. Let $y(t) = (\ x(t)\ )^3$. Use the Chain rule to find the first order derivative of $y(t)$. Show the work and answer using rigorous differential notation.

2. Let $x(a) = (\ y(a)\ )^6$. Use the Chain rule to find the first order derivative of $x(a)$. Show the work and answer using rigorous differential notation.

3. Let $x(t) = (\ g(t)\ )^6$. Use the Chain rule to find the first order derivative of $x(t)$. Show the work using rigorous differential notation.

4 - 6. Find the derivative for each equation in 4 - 6. If the Chain rule for implicit differentiation is required, show the four-step method for implicit differentiation using rigorous differential notation.

4. Let $y^2 - 2x^3 + 4 = y$. Find $\frac{dy}{dx}$.

(a) What does the quantity $\frac{dy}{dx}$ imply about x and y?______________________________________

Can we solve for y as a function of x? Yes / No

(b)

- step 1: Rewrite the equation showing the function explicitly.

- step 2: Use the Chain rule on the composite function terms and give the result in differential form.

- step 3: Rewrite the Chain rule result using the shorthand prime notation.

- step 4: Find $\frac{dy}{dx}$

5. Let $(1 + 2y)^2 = x + \tan y + 6$. Find $\frac{dy}{dx}$.

(a) What does the quantity $\frac{dy}{dx}$ imply about x and y?___________________________________

Can we solve for y as a function of x? Yes / No

Expand, simplify, and write the equation without any parentheses:

(b)

- step 1: Rewrite the equation showing the function explicitly.

- step 2: Use the Chain rule on the composite function terms and give the result in differential form.

- step 3: Rewrite the Chain rule result using the shorthand prime notation.

- step 4: Find $\frac{dy}{dx}$.

6. Let $4a^2 + 2x^3 = x + a^4 + 6$ Find $\dfrac{dx}{da}$.

(a) What does the quantity $\dfrac{dx}{da}$ imply about x and a?____________________________________

Can we solve for x as a function of a? Yes / No

Expand and write the equation without any parentheses:

(b)

- step 1: Rewrite the equation showing the function explicitly.

- step 2: Use the Chain rule on the composite function terms and give the result in differential form.

- step 3: Rewrite the Chain rule result using the shorthand prime notation.

- step 4: Find $\dfrac{dx}{da}$